THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE TWO-CHAPTER TEST SET B

Chapters 9 & 10: **Light – Reflection and Refraction, The Human Eye and the Colourful World** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. For a convex lens, Student A says its focal length is positive; Student B says negative. Then: 1

(a) Student B is correct **(b)** Student A is correct
(c) Both are correct **(d)** Both are wrong

2. The angle of incidence for a ray incident normally on a mirror is: 1

(a) 180° **(b)** 90° **(c)** 0° **(d)** 45°

3. Which mirror is used in solar furnaces? 1

(a) concave **(b)** convex **(c)** plane **(d)** none

4. Magnification $m = -2$ for a mirror means the image is: 1

(a) real, inverted and magnified **(b)** virtual and erect
(c) diminished **(d)** same size

5. Light enters glass from air at an angle; the angle of refraction is: 1

(a) greater **(b)** equal
(c) less than the angle of incidence **(d)** 90°

6. Cataract is a condition in which the: 1

(a) retina detaches **(b)** eye lens becomes milky and cloudy
(c) pupil is blocked **(d)** cornea becomes curved

7. For the colour of the sky seen at sunset, Student A says reddish; Student B says violet. Then: 1

(a) Student B is correct **(b)** Student A is correct
(c) Both are correct **(d)** Both are wrong

8. The time for which the image persists on the retina is about: 1

(a) 1/2 s **(b)** 1/16 s **(c)** 1/100 s **(d)** 1 s

9. The colour that is scattered the most by air molecules is: 1

(a) blue/violet **(b)** yellow **(c)** red **(d)** orange

10. The angle between the incident ray and the emergent ray in a prism is called: 1

(a) angle of refraction **(b)** angle of incidence
(c) angle of prism **(d)** angle of deviation

INSTEP

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 10–15 words each.

11. The power of a lens is +4 D. Find its focal length and type. 2

12. A student says that the refractive index has a unit. Justify or refute. 2

13. What is the cause of dispersion of white light? 2

14. A student says that the clear sky appears blue due to reflection from the sea. Justify or refute. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Why is the refractive index of a medium a constant? Write the relation of refractive index with speed of light. 3

16. What is the difference between the near point and far point of an eye? What are their values for a normal eye? 3

17. A convex lens forms a real image of the same size at 40 cm. Find its focal length and power. 3

OR

What are the differences between a real image and a virtual image? 3

18. Explain the Tyndall effect. Where can it be observed? 3

OR

Why does the Sun appear red at sunrise and sunset but white at noon? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) State the laws of refraction. 5

(ii) Light passes from air into a medium where its speed is 1.5×10^8 m/s. Find the refractive index.

(iii) Explain why a tank of water appears shallower than it actually is.

OR

(i) Describe image formation by a convex lens when the object is 5

(a) between F and 2F

(b) between F and O.

(ii) A 3 cm object is placed 20 cm from a convex lens of focal length 10 cm. Find the image size.

20. (i) Name the two common defects of vision. 5

(ii) Describe the causes and correction of each with diagrams.

OR

(i) What is atmospheric refraction? 5

(ii) Explain

(a) twinkling of stars

(b) advanced sunrise.

(iii) Why don't planets twinkle?

END OF QUESTION PAPER — Total: 40 Marks